CO3 ATL-2

①.

A shepherd boy gets bored tending the town's flock. To have some fun he cries out, "wolf!" even though no wolf is in sight. The villager run to protect the flock, but then get really mad when they realize the boy was playing a joke on them. One night, the shepherd boy sees a real wolf approaching the flock and calls out "wolf!" The villagers refuse to bee fooled again and stay.

Act\Pred	+	~
y	TP	FN
n	FP	TN

- positive - wolf is actually present
- negative - no wolf is present

~~wolf came be~~ no wolf no warning.

predicted that no wolf that will come and

②. Assume there are 100 images, 30 of them depict a cat, the rest do not. A machine learning model predicts the occurrence of a cat in 25 of 30 cat images. Identify TP, FP, FN, TN and calculate the Precision and Recall and F1 score.

pred \	y	no	
y	TP = 25	FP = 20	45
no	FN = 5	TN = 30	55

Total images = 100

actual cat images = 30

· Actual no-cat images = 70

· Model predicts cat in 25 of 30 cat images

recall :-

$$= \frac{25}{30} \times 100$$

$$= \cancel{0.833}\ 83.3$$

precision :-

$$= \frac{TP}{TP + FP}$$

$$= \frac{25}{25+20} = \frac{25}{45} \times 100 = 55.56$$

F1 score

$$F_1 = \frac{2 \times \text{precision} \times \text{Recall}}{\text{precision} + \text{Recall}}$$

$$= \frac{2(55.56)(83.3)}{55.56 + 83.3}$$

$$= 66.67\%$$

3. An email service provider wants to classify emails as spam or not spam using features:

- contains "offer" (Y/N)
- contains "free" (Y/N)
- Email length (S/L).

sample dataset

y, y Long Not spam

yes no long not spam

no yes short spam

no no long not spam

yes yes long spam

no no short not spam

Calculate the Prior Probabili... ...not spam)

There are 6 emails

- spam = 3
- not spam = 3

$P(spam) = \frac{3}{6} = 0.5$

$P(not\ spam) = \frac{3}{6} = 0.5$

2. Conditional probibility for each feature

For spam

offer	free	length
yes	yes	short
no	yes	short
yes	yes	long.

Therefore

$P(offer = yes | spam) = \frac{2}{3}$

$P(free = no | spam) = \frac{0}{3} = 0$

$P(length = short | spam) = \frac{2}{3}$

For not spam

not spam emails:

offer	free	length
yes	no	long
no	no	long
no	no	short

therefore

$P(offer = yes | not\ spam) = 1/3$

$P(free = no | not\ spam) = \frac{3}{3} = 1$

$P(length = short | not\ spam) = 1$

3. Naive Bayes classification

For spam:

$P(spam|x) \propto P(spam) \times P(yes|spam)$

$= 0.5 \times \frac{2}{3} \times 0 \times \frac{2}{3}$

$= 0$

for not spam

$P(notspam|x) \propto P(notspam) \times P(yes|notspam)$

$= 0.5 \times \frac{1}{3} \times 1 \times \frac{1}{3}$

$= \frac{1}{18} = 0.0556$

final classification

spam $= 0$

not spam $= 0.0556$

offer	free	length	class
yes	yes	short	spam
yes	no	long	not spam
no	yes	short	spam
no	no	long	not spam
yes	yes	long	spam
no	no	short	not spam

Aim:

Naive Bayes classifier

Attribute	class spam	not spam
offer		
yes	3.0	2.0
no	2.0	3.0
Total	3.0	5.0
free		
yes	4.0	1.0
no	1.0	4.0
Total	5.0	5.0
length		
short	3.0	2.0
long	1.0	3.0
total	5.0	5.0

graph